

# Practical Session 3: Nonlinear Systems

## Modeling and Stability

Control of Aerospace Systems

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August 10, 2026

## 1 Introduction

This report analyzes nonlinear control systems, where stability depends on initial conditions and input magnitude. We explore typical nonlinear block effects, limit cycle prediction via Describing Functions, and Phase Plane stability analysis.

## 2 Task 1: Typical Nonlinear Effects

The first step is to analyze the effect of standard nonlinearities on a pure sinusoidal input wave using Simulink.

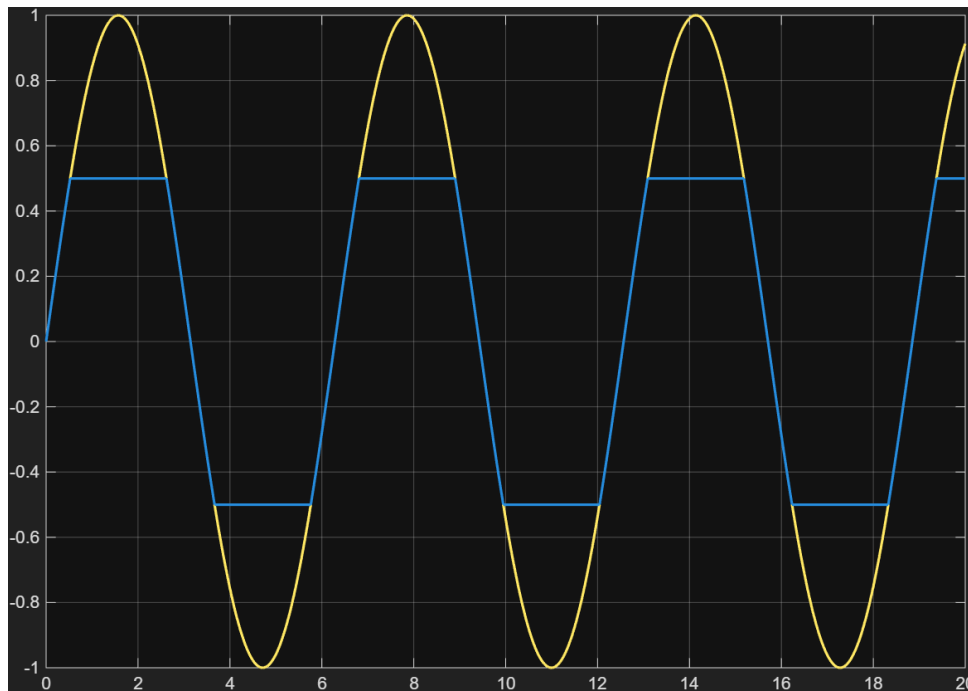


Figure 1: Time response of the sinusoidal signal passing through saturation, relay, dead zone, and backlash blocks.

As observed in Figure 1, each nonlinearity distorts the signal uniquely:

- **Saturation:** Clips the sine wave crests at predefined physical limits.
- **Relay:** Converts the input into a square wave, with hysteresis delaying the switching point.
- **Dead Zone:** Keeps the output at zero while the input is within a specific threshold.
- **Backlash:** Introduces mechanical play, flattening the output upon input direction changes.

### 3 Task 2: Limit Cycles and Nyquist Criterion

We evaluate limit cycles for a system controlled by a relay with hysteresis ( $h$ ) and amplitude  $M = 1$ . The plant transfer function is:

$$G(s) = \frac{12.5}{s^2 + 3s + 1.25} \quad (1)$$

#### 1. Nyquist Analysis

Plotting the Nyquist curve of the plant  $G(j\omega)$  and the negative inverse describing function of the relay  $-1/N(A)$  allows us to predict sustained oscillations.

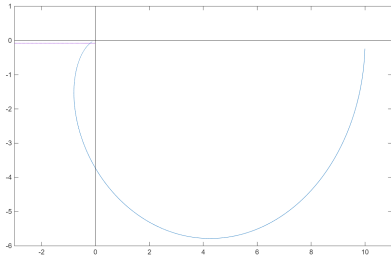


Figure 2:  $h = 0.1$

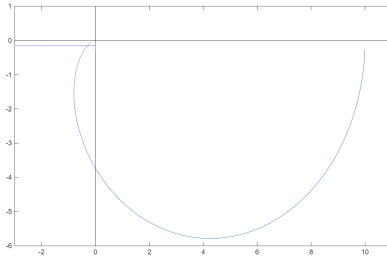


Figure 3:  $h = 0.2$

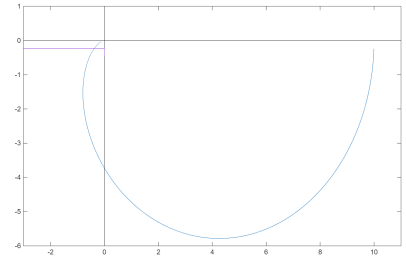


Figure 4:  $h = 0.3$

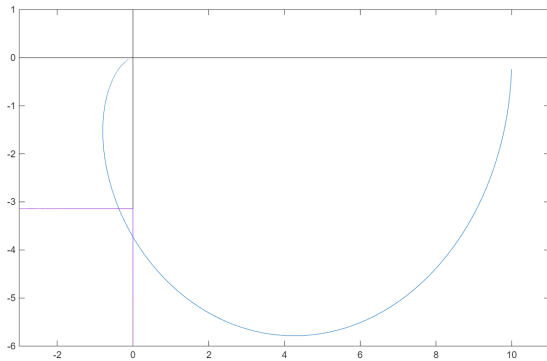


Figure 5:  $h = 4$

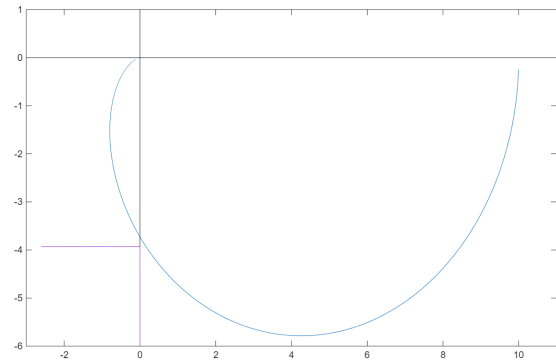


Figure 6:  $h = 5$

#### 2. Solution and Theoretical Limitations

For hysteresis values up to  $h = 4.0$ , the Describing Function locus successfully intersects the Nyquist curve, accurately predicting the existence of limit cycles. However, for  $h =$

5.0, the theoretical locus passes below the plant's curve (no intersection). Despite this, the Simulink simulation reveals a sustained limit cycle.

This discrepancy occurs because the Describing Function is a first-harmonic approximation. For large nonlinearities (severe hysteresis like  $h = 5.0$ ), higher-order harmonics become significant, rendering the basic theoretical approximation inaccurate. The actual system behavior is summarized in Table 1.

| Hysteresis ( $h$ ) | Intersection | Solution           | Amplitude ( $A$ ) | Freq. ( $\omega$ ) [rad/s] |
|--------------------|--------------|--------------------|-------------------|----------------------------|
| 0.1                | Yes          | Stable Limit Cycle | 0.26              | 7.54                       |
| 0.2                | Yes          | Stable Limit Cycle | 0.42              | 5.85                       |
| 0.3                | Yes          | Stable Limit Cycle | 0.56              | 5.02                       |
| 4.0                | Yes          | Stable Limit Cycle | 4.20              | 1.57                       |
| 5.0                | No           | Limit Cycle        | 5.15              | 1.25                       |

Table 1: Limit cycle parameters (values for  $h = 5.0$  extracted directly from simulation).

### 3. Time Response Verification

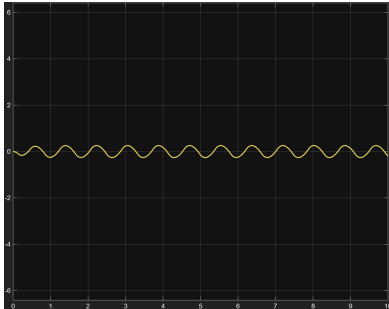


Figure 7:  $h = 0.1$

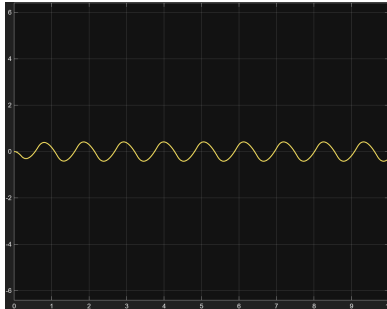


Figure 8:  $h = 0.2$

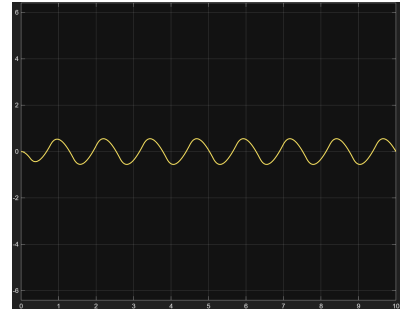


Figure 9:  $h = 0.3$

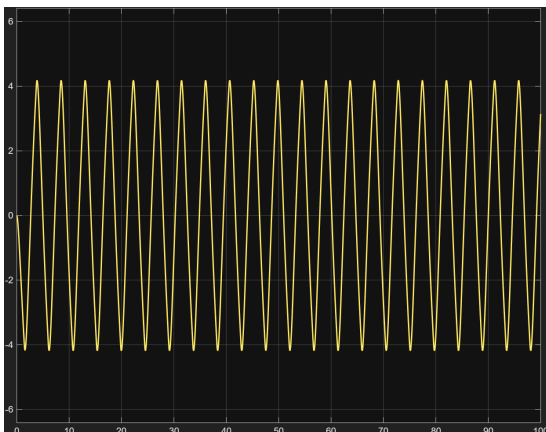


Figure 10:  $h = 4$

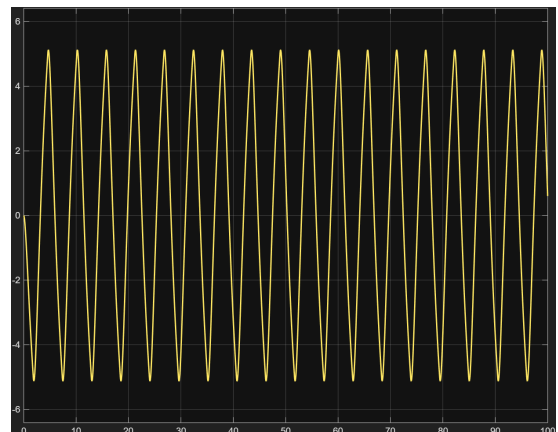


Figure 11:  $h = 5$

Figure 12: Simulink time response showing the sustained limit cycles.

Simulink time domain results demonstrate that increasing hysteresis drastically increases amplitude and reduces frequency. It also proves the existence of the limit cycle for  $h = 5.0$ ,

exposing the limitation of the Describing Function method.

## 4 Task 3: Phase Plane Analysis

To evaluate stability under different initial conditions, we define the state variables as  $x_1 = x$  and  $x_2 = \dot{x}$ .

### 4.1 System A: Van der Pol Oscillator

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -0.2(x_1^2 - 1)x_2 - x_1 \end{cases} \quad (2)$$

Trajectories from both  $(0.01, 0.01)$  and  $(5, 5)$  converge to the same closed curve, indicating a stable limit cycle.

### 4.2 System B

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -(x_2^3 + x_2)x_1 - x_1 \end{cases} \quad (3)$$

Trajectories spiral outwards, indicating the system is unstable and diverges to infinity.

### 4.3 System C

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -0.6x_2 - 3x_1 - x_1^2 \end{cases} \quad (4)$$

Stability is local. Small initial conditions like  $(0.01, 0.01)$  converge asymptotically to the origin, while larger ones like  $(4, 4)$  diverge, showing a bounded region of attraction.

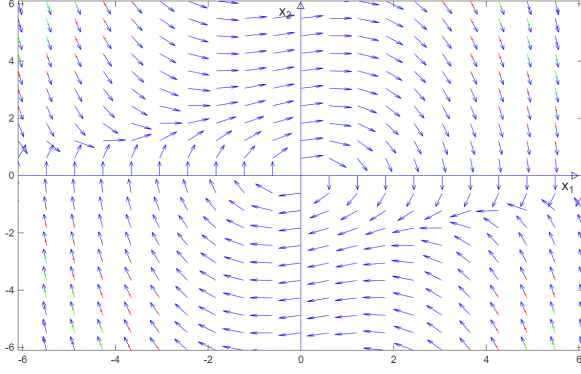


Figure 13: System A: Stable limit cycle

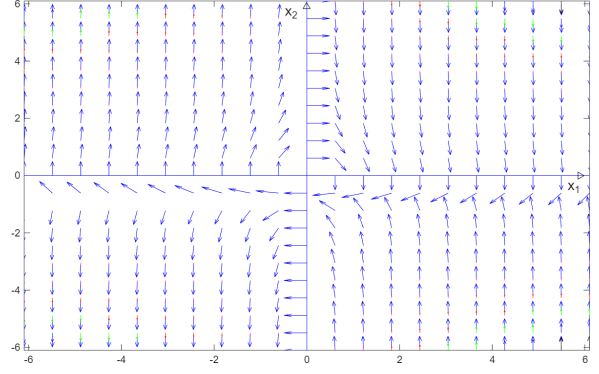


Figure 14: System B: Unstable

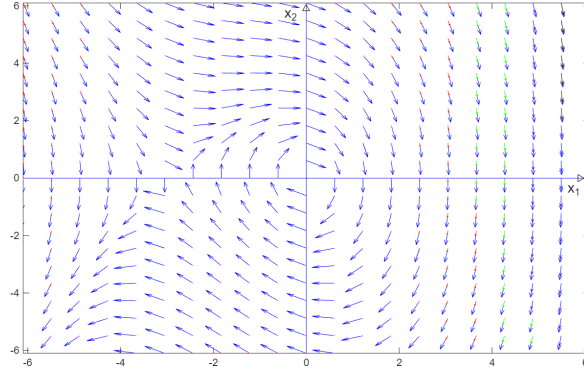


Figure 15: System C: Local stability

Figure 16: Phase plane trajectories for Systems A, B, and C.

## 5 Conclusions

Nonlinear systems exhibit unique behaviors not seen in linear ones. We observed how physical constraints distort signals (Task 1) and evaluated limit cycles (Task 2). Crucially, the analysis demonstrated that while the Describing Function method is excellent for predicting oscillations, it can fail for extreme nonlinearities (like  $h = 5.0$ ) due to its first harmonic approximation, requiring simulation to reveal the true behavior. Finally, Phase Plane analysis (Task 3) visually proved that nonlinear stability depends heavily on initial conditions, showcasing scenarios of local stability, instability, and limit cycles.